

Problem 5

Evaluate the integral:

$$I = \int \frac{\sqrt{x^4 + x^{-4} + 2}}{x^3} dx.$$

Solution

To solve the given integral, we start by simplifying the term under the square root.

Step 1: Simplify the square root

The term inside the square root is:

$$x^4 + x^{-4} + 2.$$

Notice that $x^4 + x^{-4} + 2$ can be rewritten using the identity:

$$x^4 + x^{-4} + 2 = (x^2 + x^{-2})^2.$$

Thus:

$$\sqrt{x^4 + x^{-4} + 2} = \sqrt{(x^2 + x^{-2})^2} = x^2 + x^{-2}.$$

Substituting this back into the integral, we have:

$$I = \int \frac{x^2 + x^{-2}}{x^3} dx.$$

Step 2: Simplify the integrand

Simplify the terms:

$$\frac{x^2}{x^3} = \frac{1}{x}, \quad \frac{x^{-2}}{x^3} = x^{-5}.$$

Thus, the integral becomes:

$$I = \int \frac{1}{x} dx + \int x^{-5} dx.$$

Step 3: Integrate each term

1. **First term: $\int \frac{1}{x} dx$ **

$$\int \frac{1}{x} dx = \ln|x|.$$

2. **Second term: $\int x^{-5} dx$ ** Using the power rule for integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad \text{for } n \neq -1,$$

we have:

$$\int x^{-5} dx = \frac{x^{-5+1}}{-5+1} = \frac{x^{-4}}{-4} = -\frac{1}{4x^4}.$$

Step 4: Combine the results

Combine the results of the two integrals:

$$I = \ln|x| - \frac{1}{4x^4} + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$I = \ln|x| - \frac{1}{4x^4} + C.$$